

Example

An ideal-gas mixture has the following volumetric analysis

Component	% by volume
N ₂	60
CO ₂	40

- What will be the analysis in mass basis.
- What is the mass of 1 m³ of this gas when $P = 1.5 \text{ Mpa}$ & $T = 30^\circ\text{C}$?
- Find the specific heats at 300K.
- If the gas is heated in steady-flow process & temp. increased by 120°C , find the required heat transfer.
- If the mixture undergoes an isentropic process from 0.1 Mpa , 30°C , to 0.2 Mpa , find T_2 .
- Find ΔS_m per kg of mixture, when the mixture is compressed ~~isothermally~~ from 0.1 Mpa to 0.2 Mpa .
- If both N₂ & CO₂ are supplied in separate lines at 0.2 Mpa & 300K to a mixing chamber and are mixed adiabatically. Determine the entropy change due to the mixing process per unit mass of mixture.

Solution

Assumption

- Ideal gas components & mixture.
- Steady-flow process.
- Neglect KE & PE, Work interaction.
- No heat loss.

Analysis

a. X_i

analysis

Comp.	V_i	y_i	$M_i \left[\frac{\text{kg}}{\text{kmol}} \right]$	$y_i M_i \left[\frac{\text{kg}}{\text{kmol}} \right]$	$X_i = y_i M_i / M_m \left[\frac{\text{kg}_i}{\text{kg}_m} \right]$	$R_i \left[\frac{\text{kJ}}{\text{kg}} \right]$
N ₂	0.6	0.6	28	16.8	0.488	0.2968
CO ₂	0.4	0.4	44	17.6	0.512	0.1889
$V_m = \sum_{i=1}^k V_i = 1$			$\sum y_i = 1$		$M_m = \sum y_i M_i = 34.4 \text{ kg/kmol}$	
						$\sum_{i=1}^k X_i = 1$

$$X_i = \frac{m_i}{m_m} = \frac{n_i M_i}{n_m M_m} = \frac{y_i M_i}{M_m}$$

b. $V_m = 1 \text{ m}^3$, $p = 1.5 \text{ MPa}$, $T = 30^\circ \text{C}$

$$p V_m = m_m R_m T_m \quad \leftarrow \text{Given}$$

$$\Rightarrow m_m = \frac{p V_m}{R_m T_m} \quad \leftarrow \text{Given}$$

$$= \frac{1.5 \text{ MPa} \times 1 \text{ m}^3}{(0.24168 \text{ kJ/kg} \cdot \text{K}) \times (30 + 273) \text{ K}}$$

$$= 20.48 \text{ kg}$$

$$\Rightarrow R_m = \frac{\bar{R}}{M_m} ; R_m = \sum_{i=1}^k X_i R_i = 0.488 \times 0.2968 + (0.512 \times 0.1889) = 0.24168 \text{ kJ/kg} \cdot \text{K}$$

c. Specific heats at 300K.

Table A-2	Comp.	$C_{p,i}$ (kJ/kg·K)	$C_{v,i}$ (kJ/kg·K)
	N ₂	1.039	0.743
	CO ₂	0.846	0.657

$$\Rightarrow C_{p,m} = \sum_{i=1}^k X_i C_{p,i} = (0.488 \times 1.039) + (0.512 \times 0.846) = 0.940 \text{ kJ/kg} \cdot \text{K}$$

$$C_{v,m} = C_{p,m} - R_m = (0.940 - 0.242) \text{ kJ/kg} \cdot \text{K} = 0.698 \text{ kJ/kg} \cdot \text{K}$$

Solution

Assumptions

a.

comp	y_i	M_i	$y_i M_i$	$X_i = y_i M_i / M_m$
N_2				
CO_2				

$$y_i = \frac{n_i}{n_m} = \frac{V_i}{V_m} = \frac{V_i}{V_{t,i}}$$

$$X_i = \frac{m_i}{m_m}$$

$$y_i = \frac{n_i}{n_m}$$

$$M_i = \frac{m_i}{n_i}$$

$$X_i = \frac{n_i M_i}{m_m}$$

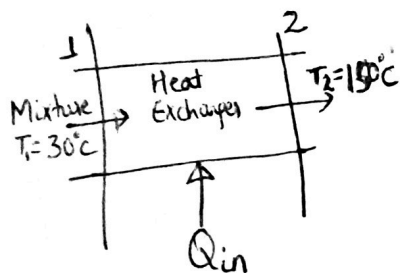
$$= \frac{n_i M_i}{m_m}$$

$$= \frac{n_i M_i}{n_m M_m}$$

$$M_m = \frac{m_m}{n_m}$$

$$X_i = y_i \frac{M_i}{M_m}$$

d.



$$\sum_{i=1}^k E_{in} - \sum_{i=1}^k E_{out} = \frac{dE_{system}}{dt}$$

$$(Q_{in} + W_{in} + \dot{m}_{in}(h_{in} + \frac{V_{in}^2}{2} + gz_{in})) - (Q_{out} + W_{out} + \dot{m}_{out}(h_{out} + \frac{V_{out}^2}{2} + gz_{out})) = 0$$

$$Q_{in} + \dot{m}_{in}(h_{in}) - \dot{m}_{out}(h_{out}) = 0$$

$$Q_{in} = \dot{m}_2 h_2 - \dot{m}_1 h_1$$

* Mass Balance:

$$\dot{m}_{in} = \dot{m}_{out}$$

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\Rightarrow \dot{Q}_{in} = \dot{m}(h_2 - h_1) = \dot{m} C_{p,m}(T_2 - T_1)$$

$$h = C_p \Delta T$$

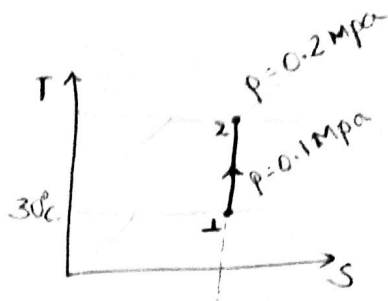
Heat transfer per unit mass; $q_{in} = \frac{\dot{Q}_{in}}{\dot{m}} = C_{p,m}(T_2 - T_1)$

$$= 0.940 \frac{kJ}{kg \cdot K} * (120) K$$

$$\underline{\underline{q_{in} = 112.8 \frac{kJ}{kg}}}$$

e. Isentropic process:
 $T_2 = ?$

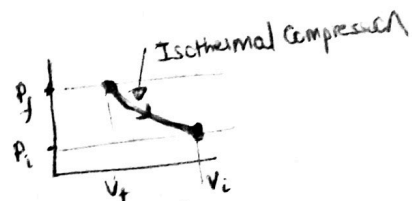
For Ideal Gas,
 Isentropic,
 Steady flow
 process



$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma_m - 1}{\gamma_m}} \quad ; \quad \gamma_m = \frac{C_{p,m}}{C_{v,m}}$$

$$\Rightarrow T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{\gamma_m - 1}{\gamma_m}} = \frac{0.940}{0.698} = 1.347$$

$$= (30 + 273) \times \left[\frac{0.2 \text{ MPa}}{0.1 \text{ MPa}}\right]^{\frac{0.347}{1.347}} = \underline{\underline{362.2 \text{ K}}}$$



f. From "e", what is ΔS_m per kg of mixture: Isothermal compression from 0.1 MPa, to 0.2 MPa

$$\Delta S_m = \frac{\Delta S_m}{m_m} = \sum_{i=1}^2 X_i \Delta S_i \quad ; \quad \Delta S_i = C_{p,i} \ln\left(\frac{T_2}{T_1}\right) - R_i \ln\left(\frac{P_{i,2}}{P_{i,1}}\right)$$

Isothermal, $T_2 = T_1$

$$\Rightarrow \Delta S_i = -R_i \ln\left(\frac{P_{i,2}}{P_{i,1}}\right)$$

Partial pressures:

$$\frac{P_i}{P_m} = y_i$$

$$\Rightarrow P_i = y_i P_m \Rightarrow P_{i,2} = y_{i,2} P_{m,2}$$

$$P_i = y_{i,1} P_{m,1}$$

Components are already
 mixed before compression process:

$$\Rightarrow y_{i,2} = y_{i,1}$$

$$\Rightarrow \Delta S_i = -R_i \ln\left(\frac{P_{m,2}}{P_{m,1}}\right)$$

$$\textcircled{x} \Delta S_{N_2} = -R_{N_2} \ln\left(\frac{P_{m,2}}{P_{m,1}}\right)$$

$$= -0.296 \frac{\text{kJ}}{\text{kg}_{N_2} \cdot \text{K}} \ln\left(\frac{0.2 \text{ MPa}}{0.1 \text{ MPa}}\right)$$

$$= -0.206 \frac{\text{kJ}}{\text{kg}_{N_2} \cdot \text{K}}$$

$$\Delta S_{CO_2} = -0.1889 \frac{\text{kJ}}{\text{kg}_{CO_2} \cdot \text{K}} \ln\left(\frac{0.2 \text{ MPa}}{0.1 \text{ MPa}}\right)$$

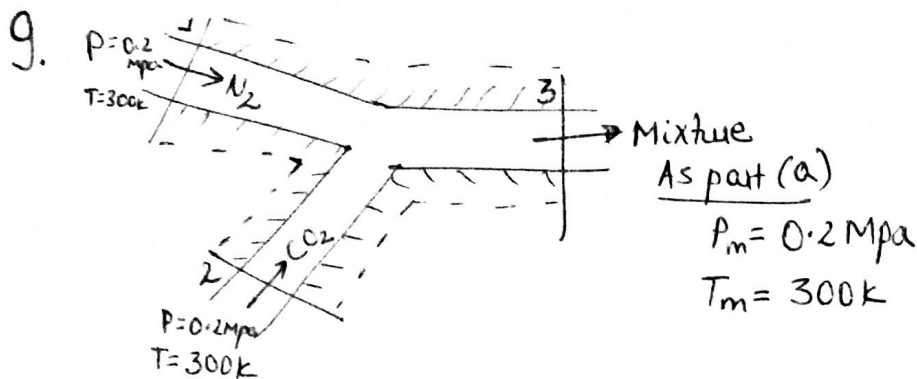
$$= -0.131 \frac{\text{kJ}}{\text{kg}_{CO_2} \cdot \text{K}}$$

$$\Delta S_m = \sum_{i=1}^2 X_i \Delta S_i = X_{N_2} \Delta S_{N_2} + X_{CO_2} \Delta S_{CO_2}$$

$$= \left(0.482 \frac{\text{kg}_{N_2}}{\text{kg}_m}\right) \left(-0.206 \frac{\text{kJ}}{\text{kg}_{N_2} \cdot \text{K}}\right) + \left(0.512 \frac{\text{kg}_{CO_2}}{\text{kg}_m}\right) \left(-0.131 \frac{\text{kJ}}{\text{kg}_{CO_2} \cdot \text{K}}\right)$$

$$\Delta S_m = -0.167 \frac{\text{kJ}}{\text{kg}_m \cdot \text{K}}$$

Becoming less disordered



Entropy change due to mixing process per unit mass of mixture.

$$\Delta S_m = \frac{\Delta S_m}{m_m} = \sum_{i=1}^2 X_i \Delta S_i$$

$$\text{And } \Delta S_i = C_{p,i} \ln\left(\frac{T_3}{T_{i,\text{initial}}}\right) - R_i \ln\left(\frac{P_{i,3}}{P_{i,\text{initial}}}\right)$$

Isothermal mixing chamber:
 $T_3 = T_2 = T_1$

Partial pressures:

$$P_i = Y_i P_m$$

$$\Rightarrow \Delta S_i = -R_i \ln\left(\frac{P_{i,3}}{P_{i,\text{initial}}}\right)$$

$$\Delta S_{N_2} = -R_{N_2} \ln\left(\frac{Y_{N_2,3} P_m}{Y_{N_2,1} P_{N_2,1}}\right)$$

$$\Delta S_{CO_2} = -R_{CO_2} \ln\left(\frac{Y_{CO_2,3} P_m}{Y_{CO_2,2} P_{CO_2,2}}\right)$$

Not mixed at state 1 & 2:
 (Each hold full potential)

$$Y_{N_2,1} = 1$$

$$Y_{CO_2,2} = 1$$

Mixture, state 3: (as part a)

$$Y_{N_2,3} = 0.6$$

$$Y_{CO_2,3} = 0.4$$

$$\Rightarrow \Delta S_{N_2} = -0.296 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \ln\left(\frac{0.6 \times 0.2 \text{ MPa}}{1 \times 0.2 \text{ MPa}}\right)$$

$$= 0.152 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$\Delta S_{CO_2} = -0.1889 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \ln\left(\frac{0.4 \times 0.2 \text{ MPa}}{1 \times 0.2 \text{ MPa}}\right)$$

$$= 0.173 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$\Delta S_m = \sum_{i=1}^2 X_i \Delta S_i$$

$$= \left(0.6 \frac{\text{kmol}}{\text{kmol}}\right) \left(0.152 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}\right) + \left(0.4 \frac{\text{kmol}}{\text{kmol}}\right) \left(0.173 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}\right)$$

$$\Delta S_m = 0.163 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Universe tends toward increased entropy
 An increase in disorder

Real Gas Mixture

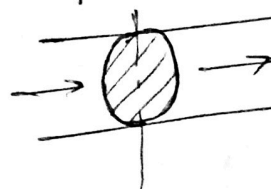
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The volumetric analysis of a mixture of gases is given.
The volumetric & mass flow rate are to be determined using three methods.

- Ideal gas relation
- Amagat's law for real gas mixture
- kay's rule
- Dalton's law for real gas mixture.

30% O₂
40% N₂
10% CO₂
20% CH₄
(By volume)

→ Decom Mixture
8 MPa, 15°C, v = 3 m/s



Solution

Given

→ Mixture
30% CO₂
40% N₂
10% CO₂
20% CH₄
@ T_m = (15 + 273) K
P_m = 8 MPa

Required: $\dot{m}$] using a, Ideal gas relation?
b, Real gas mixture using Amagat's law?
c, " " " " kay's rule?
d " " " " Dalton's law?

Assumptions

for a. Ideal gas assumptions considered for components and mixture.
- Negligible interaction

Mole analysis / Volumetric analysis

a. $\frac{PV}{m} = \frac{mRT}{m}$

$Pv = RT$

$\Rightarrow P_m v_m = R_m T_m$

$v_m = \frac{R_m T_m}{P_m}$

$v_m = \frac{0.2927 \frac{kJ}{kg \cdot K} \times (15 + 273) K}{8000 \frac{kPa}{m^2}}$

$= 0.01054 \frac{m^3}{kg}$

Volume flow rate: $\dot{V} = A \cdot v$
 $= \pi (0.02)^2 \times 3 \frac{m}{s}$
 $= 0.0009425 \frac{m^3}{s}$

$R_m = \frac{\bar{R}}{M_m}$

and $M_m = \frac{m_m}{n_m}$ or $M_m = \sum_{i=1}^k y_i M_i$

$= \frac{2840 \text{ kg}}{100 \text{ kmol}}$
 $= 28.40 \text{ kg/kmol}$

$\Rightarrow R_m = \frac{8.314 \frac{kJ}{kmol \cdot K}}{28.40 \text{ kg/kmol}}$
 $= 0.2927 \frac{kJ}{kg \cdot K}$

Comp.	$V_f(\%)$	$\frac{n_i}{n_m} = V_{fi} = y_i = \frac{n_i}{n_m} = \frac{m_i}{m_m} \times \frac{M_m}{M_i}$
O ₂	30%	$\frac{30}{100} \times 100 \text{ kmol} = 30 \text{ kmol}$
N ₂	40%	$\frac{40}{100} \times 100 \text{ kmol} = 40 \text{ kmol}$
CO ₂	10%	$\frac{10}{100} \times 100 \text{ kmol} = 10 \text{ kmol}$
CH ₄	20%	$\frac{20}{100} \times 100 \text{ kmol} = 20 \text{ kmol}$

$m_m = \sum_{i=1}^k m_i$
 $= 2840 \text{ kg}$

mass flow rate:
specific volume $\dot{V}_m = \frac{\dot{V}}{\dot{m}}$ & Total Volume flow rate

$\Rightarrow \dot{m} = \frac{\dot{V}}{v_m} = \frac{0.0009425 \frac{m^3}{s}}{0.01054 \frac{m^3}{kg}}$
 $= 0.08942 \frac{kg}{s}$

b. Amagat's Law for Real Gas:

(-2-)

Component	$T_{adu,m}(K)$	$T_{cr,i}(K)$	$T_{r,i}$	$P_{adu,m}(Mpa)$	$P_{cr,i}(Mpa)$	$P_{r,i}$
O ₂	(15+273) <u>=288</u>	154.8	1.86	8 Mpa	5.08	1.575
N ₂		126.2	2.28		3.39	2.360
CO ₂		304.2	0.947		7.39	1.083
CH ₄		191.1	1.507		4.64	1.724

$$Z_{O_2} \text{ Using } \left. \begin{array}{l} T_{r,O_2} = 1.86 \\ P_{r,O_2} = 1.575 \end{array} \right\} Z_{O_2} = 0.95$$

$$CO_2 \text{ Using } \left. \begin{array}{l} T_{r,CO_2} = 0.947 \\ P_{r,CO_2} = 1.083 \end{array} \right\} Z_{CO_2} = 0.199$$

$$N_2 \text{ Using } \left. \begin{array}{l} T_{r,N_2} = 2.28 \\ P_{r,N_2} = 2.36 \end{array} \right\} Z_{N_2} = 0.99$$

$$CH_4 \text{ Using } \left. \begin{array}{l} T_{r,CH_4} = 1.507 \\ P_{r,CH_4} = 1.724 \end{array} \right\} Z_{CH_4} = 0.85$$

$$\Rightarrow Z_m = \sum_{i=1}^4 y_i Z_i = y_{O_2} Z_{O_2} + y_{CO_2} Z_{CO_2} + y_{N_2} Z_{N_2} + y_{CH_4} Z_{CH_4}$$

$$= (0.3 \cdot 0.95) + (0.1 \cdot 0.199) + (0.4 \cdot 0.99) + (0.2 \cdot 0.85)$$

$$= \underline{0.8709}$$

$$\text{So: } v = \frac{Z_m R_m T_m}{P_m} = (0.8709) \left[\frac{(0.2927 \text{ kPa} \cdot \text{m}^3 / \text{kg} \cdot \text{K}) \cdot 288 \text{ K}}{8000 \text{ kPa}} \right]$$

$$= 0.009178 \text{ m}^3/\text{kg}$$

$$\text{And } \dot{V} = 0.0009425 \text{ m}^3/\text{s}$$

$$\dot{m} = \frac{\dot{V}}{v} = \frac{0.0009425 \text{ m}^3/\text{s}}{0.009178 \text{ m}^3/\text{kg}} = \underline{0.10269 \text{ kg/s}}$$

$$J = F \cdot S = N \cdot m = P \cdot A \cdot S = P_a \cdot m^3$$

$$\% \text{ error} = \frac{(V_{m,ideal} - V_{m,amagat})}{V_{m,amagat}} \cdot 100$$

$$= \frac{(0.01054 - 0.009178)}{0.009178} \cdot 100$$

(-3-)

C. Kay's Rule

Component	Y_i	$T_{cr,i}$	$Y_i * T_{cr,i}$	$\sum_{i=1}^4 Y_i T_{cr,i} = T'_{cr,m}$	$P_{cr,i}$	$Y_i * P_{cr,i}$	$\sum_{i=1}^4 Y_i P_{cr,i} = P'_{cr,m}$
O ₂	0.3	154.8		165.6 K	5.08		4.547 Mpa
N ₂	0.4	126.2			3.39		
CO ₂	0.1	304.2			7.39		
CH ₄	0.2	191.1			4.64		

$$\left. \begin{aligned} \text{Then } T_R &= \frac{T_{\text{actual},m}}{T'_{cr,m}} = \frac{288 \text{ K}}{165.6 \text{ K}} = 1.739 \\ P_R &= \frac{P_{\text{actual},m}}{P'_{cr,m}} = \frac{8 \text{ MPa}}{4.547 \text{ MPa}} = 1.759 \end{aligned} \right\} Z_m = 0.92$$

$$\Rightarrow v = Z_m \left[\frac{R T_m}{P_m} \right] = \text{---} = 0.009694 \text{ m}^3/\text{kg}$$

$$\dot{V} = 0.0009425 \text{ m}^3/\text{s}$$

$$\% \text{ Error} = (0.01054 - 0.009694) \times 100$$

$$\Rightarrow \dot{m} = \frac{\dot{V}}{v} = \frac{0.0009425 \text{ m}^3/\text{s}}{0.009694 \text{ m}^3/\text{kg}} = \underline{\underline{0.009723 \text{ kg/s}}}$$

(-4-)

d. Dalton's Law for Real Gas Mixture:

@ T_m, V_m ; but V_m is unknown: SO Iteration will be used.

component	R_i ($\frac{m^3}{kg \cdot K}$)	$T_{cr,i}$ (K)	$P_{cr,i}$ (kPa)	$V_{cr} = \frac{R_i T_{cr,i}}{P_{cr,i}}$	$V_{m,ideal}$	$V_{m,1^{st} iteration}$	$V_R = \frac{V_m}{V_{cr}}$	T_R	$V_{R(2)}$
O ₂	0.2598	154.8	5080	0.0079	0.01054 $\frac{m^3}{kg}$		1.33	1.86	
N ₂	0.2968	126.2	3390	0.0110			0.958	2.28	
CO ₂	0.1889	304.2	7390	0.00777			1.35	0.947	
CH ₄	0.5182	191.1	4640	0.02134			0.494	1.507	
							Z_{O_2}	Z_{N_2}	Z_{CO_2}
							Z_{CH_4}		

$$\Rightarrow Z_m = \sum_{i=1}^4 y_i Z_i = y_{O_2} Z_{O_2} + y_{N_2} Z_{N_2} + y_{CO_2} Z_{CO_2} + y_{CH_4} Z_{CH_4}$$

$$V_{m(1)} = Z_m \left[\frac{R_m T_m}{P_m} \right]$$

$$m = \frac{\dot{V}}{v} =$$

2nd iteration

Use $V_{m(1)}$
 $\rightarrow V_R$